Dr Ramachandran Thiruvengadam

M.B.,B.S., DCH, M.D. (Biochemistry), Ph.D. (Maternal and Child Health)
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Summary

Physician-scientist in maternal and child health. Develops and validates predictive and diagnostic tools and interventions deployable at public-health scale to reduce preterm birth, fetal growth restriction and the great obstetrical syndromes. Investigator in the GARBH-Ini pregnancy cohort programme (DBT); coordinated the clinical component of the DBT Consortium for COVID-19 Research (2020-2023). PI of two ICMR NHRP projects; Co-PI of the ICMR GRASPP trial and MOMI-India Phase II. Four Indian patent applications (three awarded).

Education

Institution	Degree	Year
Translational Health Science and Technology Institute, Faridabad (affiliated to Jawaharlal Nehru University	Ph.D. (Life Sciences - Maternal and Child Health)	2023
All India Institute of Medical Sciences (AIIMS), New Delhi	M.D. (Biochemistry)	2016
Institute of Child Health and Hospital for Children, Madras Medical College	Diploma in Child Health (DCH)	2009
Madurai Medical College, Madurai	M.B.,B.S.	2005

Positions

Institution	Position	From	To
BRIC-Translational Health Science and Technology Institute (THSTI), Faridabad	Associate Professor, Maternal and Child Health Programme	30 June 2025	Till date
Pondicherry Institute of Medical Sciences, Puducherry	Assistant Professor, Department of Biochemistry	May 2022	June 2025
THSTI, Faridabad	Principal Scientist - I	November 2021	May 2022
THSTI, Faridabad	Senior Research Scientist	July 2021	November 2021
THSTI, Faridabad	Scientist D	January 2020	June 2021
THSTI, Faridabad	Scientist C	January 2019	January 2020

Current grants

Project	Agency	Role	Amount
IPD analysis of hypothyroidism in pregnancy	Indian Council of Medical Research (ICMR), National Health Research Priority	PI	INR 24,51,020 (Year 1 sanctioned)
Secondary data analysis of gestational weight gain and maternal-neonatal outcomes in Indian cohorts	Indian Council of Medical Research (ICMR), National Health Research Priority	PI	INR 19,07,060 (Year 1 sanctioned)
Surveillance and Pregnancy/Perinatal Research Platform for India	Christian Medical College (CMC) Vellore	PI	INR 1,14,88,500 (excl. GST)
Graded Risk-based Assessment for Screening and Prevention of Placental Disorders	Indian Council of Medical Research (ICMR)	Co-PI	INR 9,94,20,000 (994.20 lakh)
MOMI-India	Bill & Melinda Gates Foundation, via Grand Challenges India-BIRAC	Co-PI	INR 10,65,90,000 (1065.90 lakh)
Improving maternal and neonatal outcomes using imaging data science	ICMR, Extramural Research Programme (Intermediate Grant)	Co-PI (clinical investigation)	INR 2,06,22,808 (Year 1 sanctioned); INR 7,97,46,000 proposed for 4 years

Patents

- Garbh-Ini Ultrasound Image-based Gestational Age Estimation System (GAUGE) (Indian application 202411027837; filed 4 april 2024; awarded)
- A population-specific gestational dating system and a method thereof (Indian application 202341061748; filed 13 september 2023)
- Potently neutralising novel human monoclonal antibodies to SARS-CoV-2 (COVID-19) (Indian application 202111038519; filed 25 august 2021; awarded)
- Immunoassay for detection of Coronaviridae (Indian application 202011039696; filed 14 september 2020; awarded)

Selected publications

1. Thiruvengadam R, Ayushi, Murugesan DR, et al; GARBH-Ini Study Team. Incidence of and risk factors for small vulnerable newborns in north India: a secondary analysis of a prospective pregnancy cohort. *Lancet Glob Health*. 2024;12(8):e1261-e1277. doi:10.1016/S2214-109X(24)00212-2 *[First author]*

2. Gadekar VP, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Development and external validation of Indian population-specific Garbhini-GA2 model for estimating gestational age in second and third trimesters. *Lancet Reg Health Southeast Asia*. 2024;25:100362. doi:10.1016/j.lansea.2024.100362 *[Corresponding author]*

3. Thiruvengadam R, Awasthi A, Medigeshi G, et al; DBT India Consortium for COVID-19 Research. Effectiveness of ChAdOx1 nCoV-19 vaccine against SARS-CoV-2 infection during the delta (B.1.617.2) variant surge in India: a test-negative, case-control study and a mechanistic study of post-vaccination immune responses. *Lancet Infect Dis*. 2022;22(4):473-482. doi:10.1016/S1473-3099(21)00680-0 *[First author]*

4. Thiruvengadam R, Rizvi ZA, Raghavan S, et al. Clinical and experimental evidence suggest omicron variant of SARS-CoV-2 is inherently less pathogenic than delta variant independent of previous immunity. *Eur J Med Res*. 2023;28(1):421. doi:10.1186/s40001-023-01373-3 *[First author]*

5. Chanian R, Thiruvengadam R, Desiraju BK. Prediction of spontaneous preterm birth in the second trimester: current models, limitations, and the way forward. *Indian J Pediatr*. 2026;93(8):852-858. doi:10.1007/s12098-026-06030-1 *[Second author]*

6. Dinakaran A, Thiruvengadam R, Srinivasan AR, et al. Prediction models for gestational diabetes mellitus: diagnostic utility of clinical and biochemical markers. *Indian J Endocrinol Metab*. 2025;29(3):295-302. doi:10.4103/ijem.ijem_497_24 *[Second author]*

7. Bhattacharjee E, Thiruvengadam R, Varshney D, et al; GARBH-Ini Study Group. Deciphering mechanisms leading to preterm birth through post-GWAS integration of multi-omics data. *iScience*. 2025;28(11):113861. doi:10.1016/j.isci.2025.113861 *[Second author]*

8. Bhattacharjee E, Thiruvengadam R, Ayushi, et al; GARBH-Ini Team. Genetic variants associated with spontaneous preterm birth in women from India: a prospective cohort study. *Lancet Reg Health Southeast Asia*. 2023;14:100190. doi:10.1016/j.lansea.2023.100190 *[Second author]*

9. Sebastião YV, Thiruvengadam R, Khanam R, et al. Association between gestational weight gain and adverse pregnancy outcomes: cohort analysis from South Asia and sub-Saharan Africa. *BMJ Public Health*. 2025;3(1):e000900. doi:10.1136/bmjph-2024-000900 *[Second author]*

10. Thiruvengadam R, Desiraju BK, Natchu UCM, et al; GARBH-Ini Study Team. Gestational weight gain trajectories in GARBH-Ini pregnancy cohort in North India and a comparative analysis with global references. *Eur J Clin Nutr*. 2022;76(6):855-862. doi:10.1038/s41430-021-01040-y *[First author]*

Service and recognition

- Associate Fellow, Indian National Science Academy (INSA), New Delhi (2026)
- Expert member, Project Selection Committee (Epidemiology), ICMR intramural funding, 2024 onwards.
- Academic Editor, PLOS Global Public Health, and Associate Editor, Frontiers in Pediatrics, 2023 onwards.
- Member, Technical Advisory Group, Obesity Individual Participant Data Meta-analysis, ICMR (Division of Reproductive, Child Health and Nutrition), 2026 onwards.
- Dr M.K. Bhan Group Award for the most impactful collaboration, THSTI (as a member of the clinical team of the DBT Consortium for COVID-19 Research) (2020)
- Early Career Medical Research Award, THSTI, DBT, Government of India (maternal and child health) (2017)

h-index 22 (Google Scholar, as on 5 November 2026); citations 3,956 (Google Scholar, as on 5 November 2026).